

Outset · AI in Healthcare · Day 3 · Group 5

Predicting heart disease from a checkup

A tabular model, and the fairness question at its center: does it work for women?

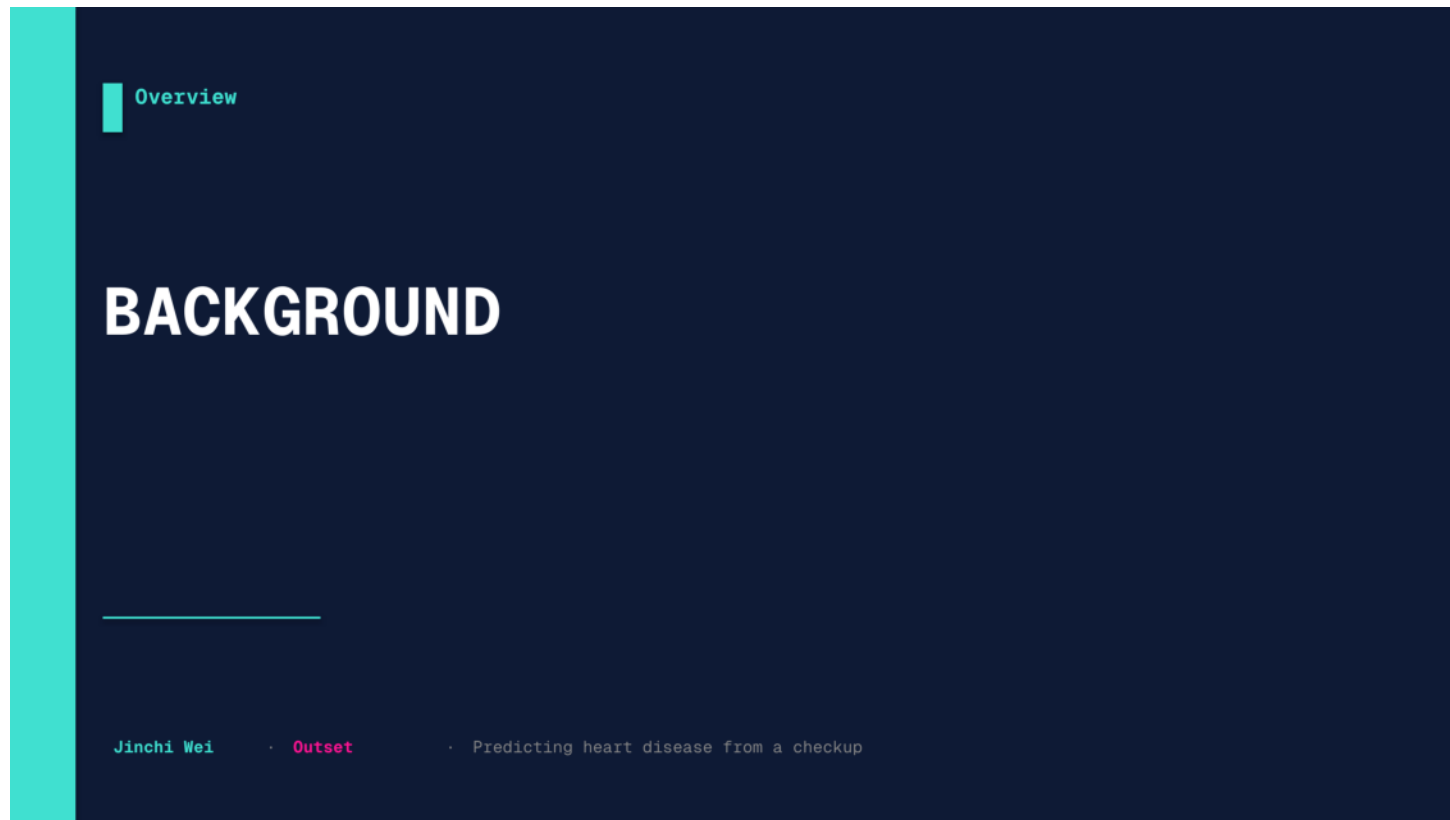
Jinchi Wei

Outset

2026-07-08



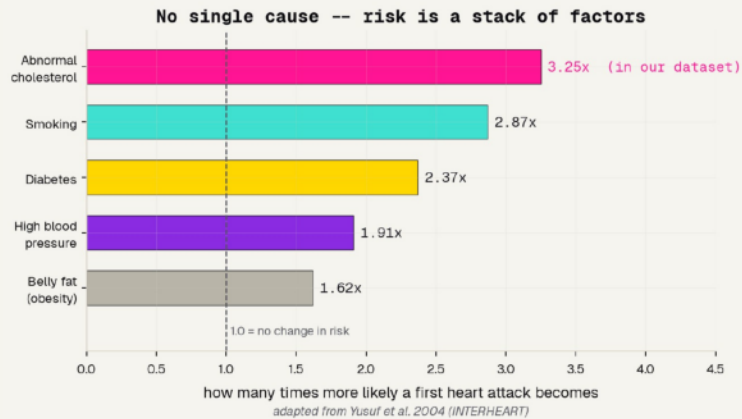
Slide 1 -



Slide 2 – Background

This first section sets up the medical problem before any code. Two facts matter: heart risk is built from many measurements at once, and women's heart disease is systematically missed. Both will shape how we read our model later.

How doctors actually estimate heart risk



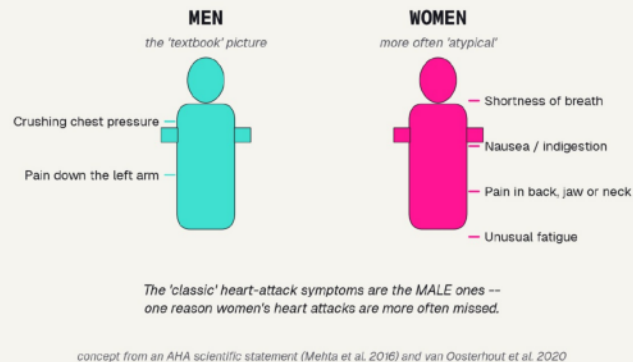
- No doctor diagnoses heart disease from one number
- They combine several routine checkup measurements into a single risk estimate, the way the Framingham Risk Score does. Worldwide, about nine risk factors led by cholesterol and smoking explain over 90% of first heart attacks. Risk is a stack of factors, not a single cause.

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Slide 3 – How doctors actually estimate heart risk

The takeaway is that no single measurement decides heart risk -- doctors add up many factors, the way the Framingham score does. The bars show how much each factor multiplies risk; cholesterol and smoking lead, but none acts alone. Hold this 'stack of factors' idea for when we test cholesterol by itself.

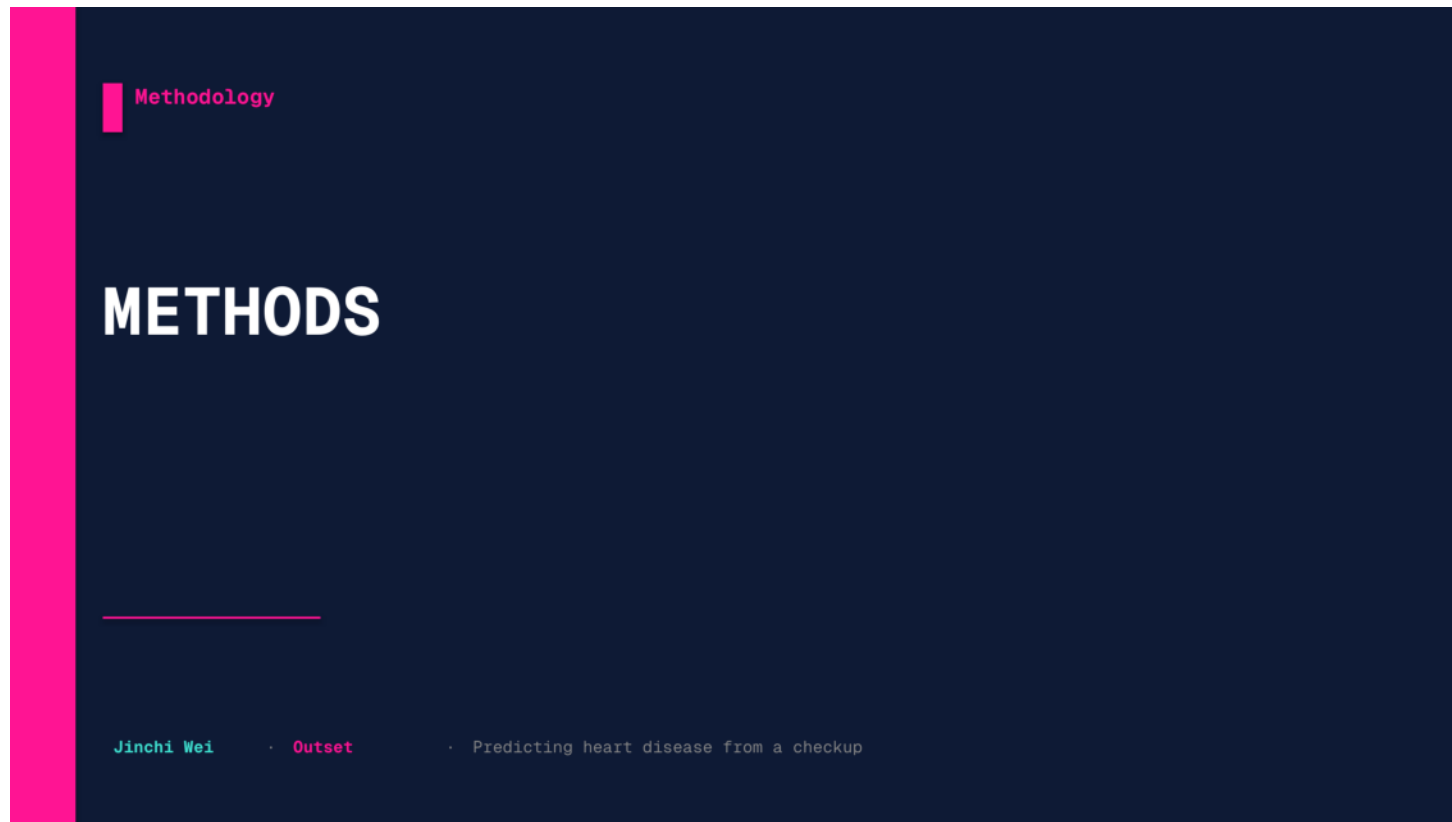
Women's heart disease is under-diagnosed



- Here is the problem that drives our project
- The "textbook" heart attack, crushing chest pain down the left arm, is the male picture. Women more often feel back pain, nausea, and shortness of breath, so their heart attacks get missed. The American Heart Association documents that women are misdiagnosed and under-treated more than men.

Slide 4 – Women's heart disease is under-diagnosed

This slide is the motivation for the whole project. The classic chest-pain picture is the male one, so women's often different symptoms get missed, and the AHA has documented that women are under-diagnosed and under-treated. That real-world bias is exactly what we will look for inside our own model.



Slide 5 – Methods

Now we turn to what we actually built. This section covers the prediction task, the dataset, and the four models we compared -- and why the choice of algorithm turns out to be the least interesting decision of all.

The task and the data

303 patients

The Cleveland Heart Disease dataset, from a 1980s study.

13 features

Age, sex, blood pressure, cholesterol, chest-pain type, exercise-ECG results.

Predict disease

1 = has heart disease, 0 = does not. Graded on held-out patients.

Slide 6 – The task and the data

Our job is a simple yes/no prediction from a routine checkup, using the classic Cleveland dataset: 303 patients, 13 features, graded on patients the model never saw. Because tabular data like this is easy for any model, the real questions are which features matter and who the model works for, not which algorithm wins.

Four models, one honest test

Logistic Regression

Simple, explainable, and here the best of the four.

Random Forest & CatBoost

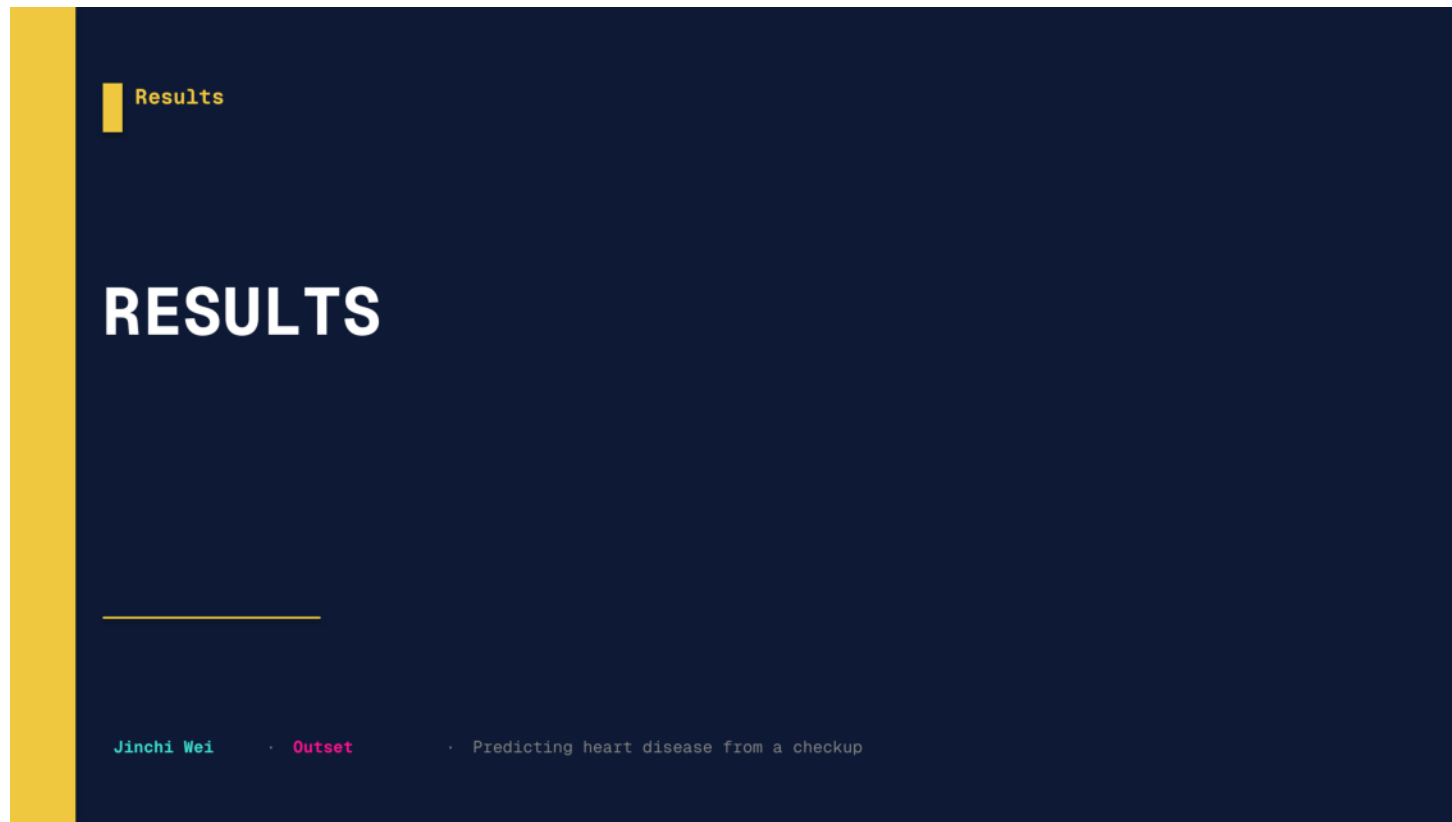
Tree ensembles, the workhorses of tabular data.

TabPFN

A foundation model for tables, like the one from Day 2.

Slide 7 – Four models, one honest test

We trained four different models -- a simple logistic regression, two tree ensembles, and a tabular foundation model -- and always compared them to the 'just guess the majority' baseline of about 0.54. Anything that cannot beat that baseline is useless. They all beat it, and they all basically tie.

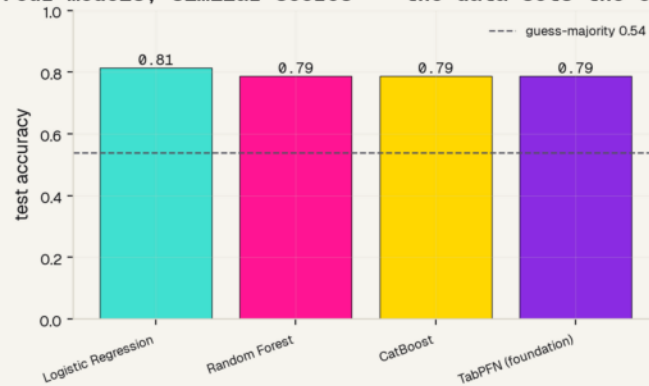


Slide 8 – Results

Here are the three findings. Watch for a pattern: the flashy model comparison teaches the least, while the two 'boring' experiments -- one feature, and a fairness split -- teach the most.

The data sets the ceiling

Four models, similar scores -- the data sets the ceiling



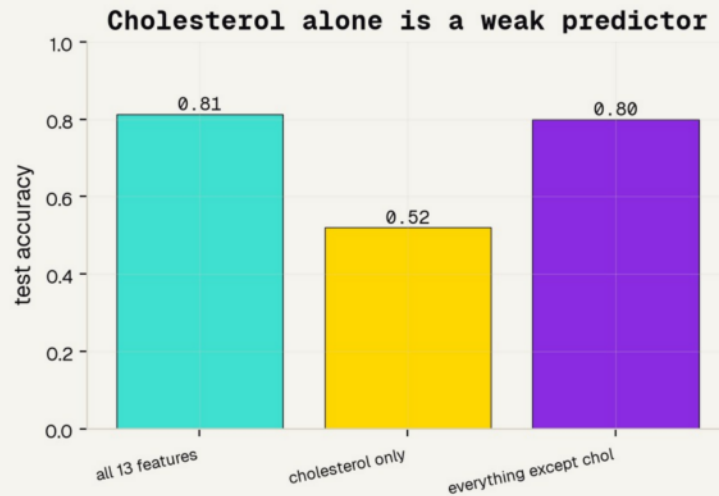
All four models land between 0.79 and 0.81, comfortably above the coin-flip baseline, and they basically tie each other. That is the lesson: on clean tabular data the model barely matters. Swapping a plain logistic regression for a fancy foundation model buys almost nothing. Three hundred patients from the 1980s contain only so much signal, and no model can invent more.

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Slide 9 – The data sets the ceiling

All four models land within a whisker of each other, around 0.80. The lesson is that on clean tabular data the algorithm barely matters -- a plain logistic regression matches a fancy foundation model. The ceiling is set by the data itself, not by how clever the model is.

Cholesterol alone is a weak predictor



Everyone “knows” cholesterol causes heart disease, so we gave a model only the cholesterol column. It scored about 0.52, basically a coin flip. Meanwhile the checkup with cholesterol removed scored about 0.80. Cholesterol is a major cause of heart disease over decades, but on its own it is a poor clue for who has disease right now. That is why doctors combine many measurements.

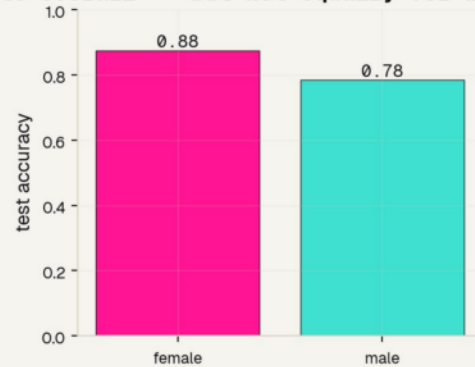
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Slide 10 – Cholesterol alone is a weak predictor

We gave one model only the cholesterol column and it scored near a coin flip, while dropping cholesterol entirely barely hurt. Cholesterol matters over decades, but on its own it is a weak clue for who has disease right now -- which is why doctors combine many measurements. This is the 'stack of factors' idea made concrete.

Accurate overall, but not equally by sex

Accurate overall -- but not equally for women and men

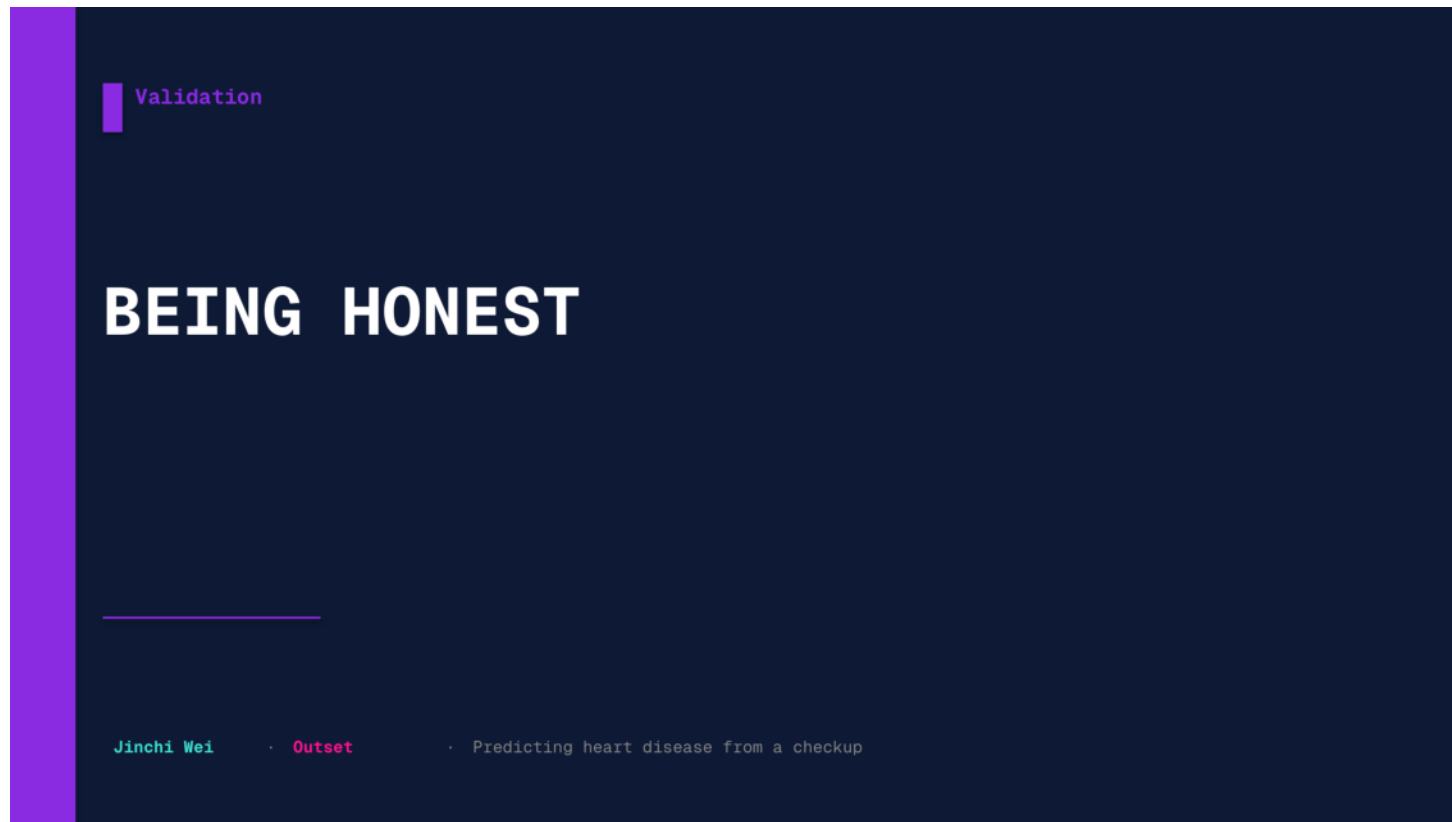


■ This is the point of the project

- One accuracy number can hide unfairness. When we grade the model separately for women and men, it is about 0.88 accurate for women and 0.78 for men, a gap of about 0.09. An accuracy gap in either direction means the model behaves differently depending on your sex. You only see it because you looked.

Slide 11 – Accurate overall, but not equally by sex

This is the heart of the project. A single accuracy number looked fine, but grading women and men separately reveals about a nine-point gap. The point is not the direction of the gap -- it is that you only discover unfairness when you deliberately look for it.

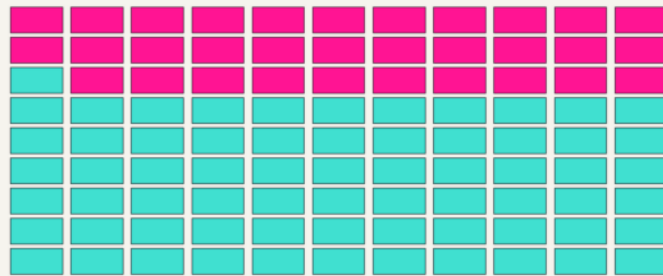


Slide 12 – Being honest

Good medical-AI work states loudly what it cannot claim. This section covers where our result comes from -- a small, skewed, decades-old dataset -- and what we are and are not allowed to conclude from it.

The data is small and male-skewed

Who is in the data? Each square = 3 patients



■ men -- 201 patients (about 2 of every 3) ■ women -- 96
from the Cleveland Heart Disease data (Detrano et al. 1989)

- A model can only learn from who is in its data, and ours is about 2:1 male
- With far fewer women, the model gets less practice on them, and the way patients were selected in the 1980s made the male rows sicker. That skew is not a bug we can fix with a better model, it is a property of the data, and it is exactly the bias that makes heart disease harder to catch in women.

Slide 13 – The data is small and male-skewed

Our data is about two men for every woman, so the model simply gets less practice on women. That skew is not a bug we can code away -- it is baked into how the 1980s study was collected, and it mirrors the same real-world bias that makes women's heart disease harder to catch. Every result above must be read through this limitation.

What this project can and cannot say

It can

Show that checkup features carry real signal, that no single number is the whole story, and that accuracy can be unequal across sex.

It cannot

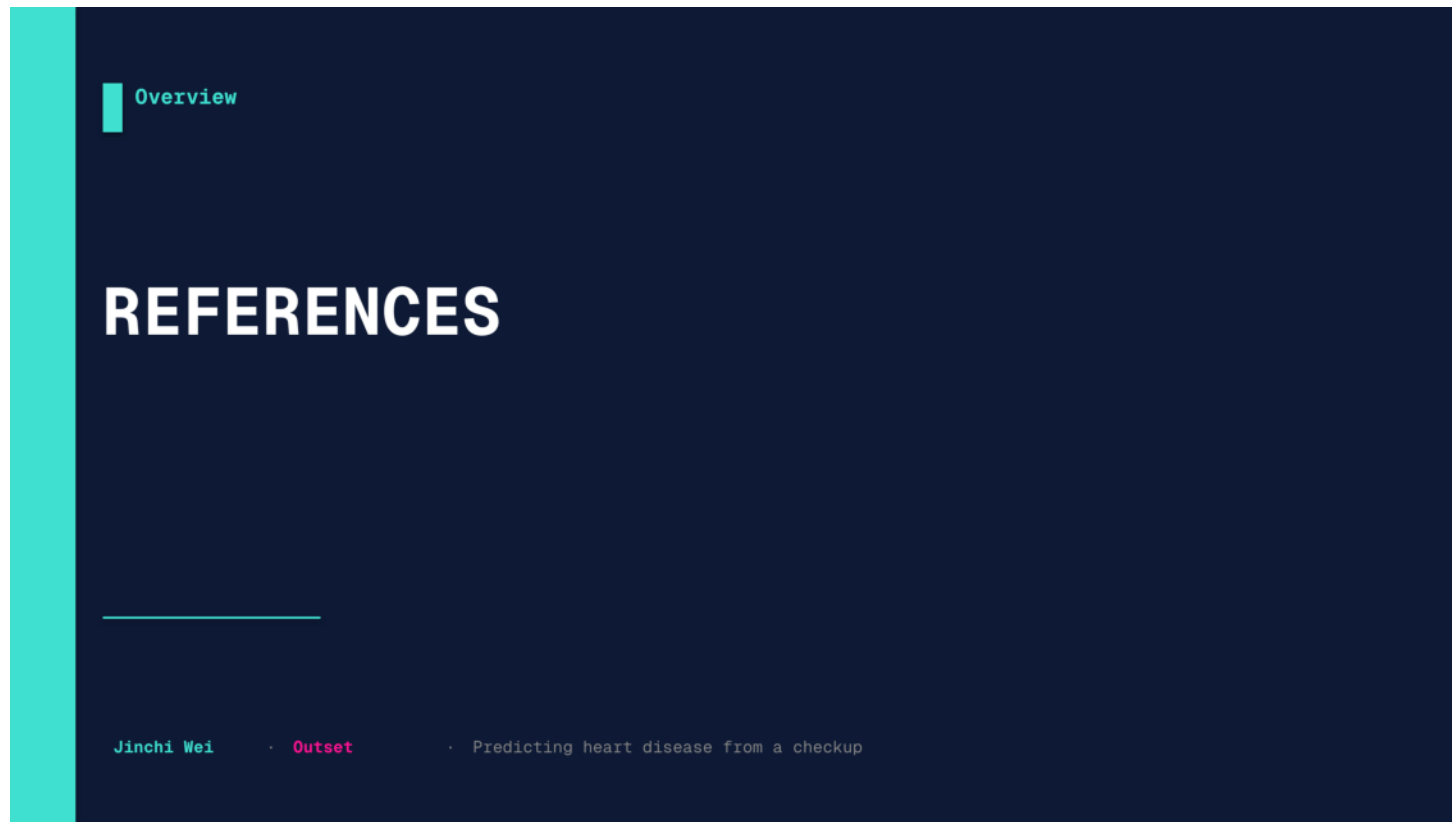
Say anything trustworthy about today's patients. This is 303 people from one 1980s study.

Watch the metric

Missing a real heart attack is far worse than a false alarm. A deployed tool would be tuned for sensitivity, not raw accuracy.

Slide 14 – What this project can and cannot say

This is the most important slide in any medical-AI talk. Our toy model earns the right to say checkup features carry signal, no single number is the whole story, and accuracy can be unequal -- but it cannot say anything trustworthy about today's patients. And in a real tool, missing a heart attack is far worse than a false alarm, so you would tune for sensitivity, not raw accuracy.



Slide 15 – References

Every claim in this deck traces to a peer-reviewed source. These are the papers behind the risk factors, the under-diagnosis of women, and the dataset itself.

Sources

Cardiac risk scoring

Wilson 1998, Framingham Risk Score, Circulation. Yusuf 2004, the INTERHEART study, Lancet.

Under-diagnosis in women

Mehta 2016, AHA scientific statement, Circulation. van Oosterhout 2020, JAHA. Bugiardini and Bairey Merz 2005, JAMA.

The dataset

Detrano 1989, American Journal of Cardiology. UCI Machine Learning Repository, Heart Disease Data Set.

Slide 16 – Sources

These are the primary references. The risk-scoring and INTERHEART work back the 'stack of factors' framing; the AHA statement and its follow-ups document the under-diagnosis of women; and Detrano 1989 is the origin of the Cleveland dataset we used.

Thank You

魏晉祺

Jinchi Wei

Outset

jinchikwei@gmail.com

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Slide 17 -